

| Field        | Description                                                    |
|--------------|----------------------------------------------------------------|
| Case ID:     | CS-2025-003                                                    |
| Case Type:   | Email messages containing malicious URL removed after delivery |
| Reported by: | Windows Defender                                               |
| Analyst:     | Ali YILMAZ                                                     |
| Date:        | 2025-10-16                                                     |
| Severity:    | Medium                                                         |
| Status:      | Closed                                                         |
| Environment: | Production                                                     |

### Case Title

Email messages containing malicious URL removed after delivery

### Case ID

Email messages containing malicious URL removed after delivery - T1566.002 - Phishing: Spearphishing Link

### Detection Source

- Microsoft Defender for Office 365 - Post-delivery remediation alert: *Email messages containing malicious URL removed after delivery*
- Alert fields: sender, recipients, subject, message ID, removed URL(s), timestamp.

### Incident Summary

Microsoft Defender detected and automatically removed post-delivery redirect links in multiple delivered emails. The links redirected through link.mail.beehiiv.com and contained a high-entropy encoded token. Local Base64 decode in CyberChef produced a binary/garbled blob (high entropy) — likely an opaque tracking token or encrypted payload. Final landing page was not recoverable by single-step decoding; safe redirect tracing / sandbox detonation required to confirm maliciousness. Risk: **Medium (pending sandbox result)**.

### MITRE ATT&CK

T1566.002 - Phishing: Spearphishing Link

### Investigation Steps

1. Review Alert Details

- Open the Defender alert titled “*Email messages containing malicious URL removed after delivery.*”
- Note sender, recipient list, subject, delivery time, and URLs that were removed.
- Confirm post-delivery remediation status “Removed.”

## 2. Extract and Analyze URL

- Copy the full redirect link `https://link.mail.beehiiv.com/...` from the alert.
- Paste into **CyberChef** and apply From Base64 (URL Safe) Detect File Type.
- Output shows **high-entropy binary data**, not readable indicates encrypted or tokenized redirect.

## 3. Check Redirect Chain Safely

- In an isolated VM (no internet access to production network), run:
- `curl -I -L -s -S "https://link.mail.beehiiv.com/..."`
- Review Location: headers to identify the **final landing domain** (do not open directly in browser).

## 4. Submit for Sandbox Analysis

- Upload the URL to a safe detonation platform (e.g., **URLScan.io**, **Any.Run**, or internal sandbox\*\*).
- Record behavior: redirected domains, page type (login, download, ad, etc.), network indicators.

## 5. Check Other Deliveries

- In Defender **Email & Collaboration > Explorer** search for messages containing `link.mail.beehiiv.com`.
- List all recipients and message subjects with similar patterns in the past 30 days.

## 6. OSINT Enrichment (if final host known)

- Query **VirusTotal**, **AbuseIPDB**, or **FortiGuard** for the final domain/IP.
- Note if reputation = *malicious / phishing / compromised site*.

## 7. Validate Containment

- Ensure all detected messages are quarantined or removed.

- Check if users clicked the link (from Defender telemetry or proxy logs).

### Outcome:

Encoded Beehiiv redirect link analyzed. No readable data from token; sandbox analysis pending. Defender successfully removed delivered messages. Incident remains **Medium severity** until final confirmation of the destination site.

## Evidence

### 1. Defender Alert Screenshot

- Title: *"Email messages containing malicious URL removed after delivery."*
- Shows affected users, sender address, and URL removed.
- Post-delivery remediation status:  *Removed successfully.*

### 2. URL Sample

`https://link.mail.beehiiv.com/ss/c/u001.i9xsYZJOglGGEQlwC6GCrPUdjVwQH-Z5qjF-ZxIOrX4Q6DR8LPHKHcRVL_e2EURxte4-y7xNBH07upiG2RV55wEi5Qkr1fK5Rm3NANp1J8udCnN-U78bjfeNtrUwX8RQZsovMzAlMLwwkWcmbZ0P_uocfpnITgPu2GbfvioUmoISNI1s_Y9D6jDbYZvA7m9uQOmnCsKaYqHAz0L7g4iypG0aZCwGZRLXVVD-uW53MML5kryXrpTHgfZdYLZtT5MZG-mpIQ5ZdB2XA6BOGunR5oWh7wGwnG1MrKhXeqjV71FM8x1BWSMz4sV05VzQsfJI7VbUD3eM_sL0nU0pWOHym5a7yv84s7nSgoDWldEylexV56h4vJFU5KVwBDxUB/...`

- Domain: link.mail.beehiiv.com
- Encoded path: Base64 (URL-safe format).

### 3. CyberChef Analysis Output

- Operation: From Base64 (URL Safe)
- Result: Non-readable binary blob (high entropy).
- Interpretation: token likely **encrypted or server-decoded**.

### 4. Redirect Chain Verification

- Command executed in isolated VM:
- `curl -I -L -s -S "https://link.mail.beehiiv.com/..."`
- Observed multiple Location: headers indicates redirect mechanism.
- Final URL not directly exposed in local.

### 5. OSINT Cross-Checks

- VirusTotal domain reputation: benign / suspicious (based on result).
- AbuseIPDB No abuse reports or low confidence.

## 6. Supporting Screenshots

- Defender alert summary
- CyberChef result window
- Curl redirect output
- Sandbox report screenshot
- Mail trace result.

## Recommended Actions

### Containment

- Confirm all affected emails were removed by Defender post-delivery remediation.
- If any copies remain, manually quarantine or delete them from user mailboxes.

### Verification / Hunting

- In **Microsoft 365 Defender Explorer**, search for other messages containing the domain link.mail.beehiiv.com.
- Review sender address, subject line, and recipient list to check for similar campaigns.
- Verify whether any users clicked the URL (Defender telemetry or proxy logs).

### User Awareness

- Notify recipients that an email with a suspicious link was removed.
- Advise users **not to click** similar newsletter-style links from unknown senders.

### Sandbox Validation

- Submit the URL to a safe detonation platform (URLScan, Any.Run, or internal sandbox).
- If final landing page shows phishing or malware content → escalate to Tier-2 for full investigation.

### Blocking / Monitoring

- If the final URL or domain is confirmed malicious:
  - Add domain/IP to mail and web proxy blocklists.
  - Enable Defender safe links policy for additional protection.

- Continue monitoring mail traffic for similar redirect domains.

### **Post-Incident Steps**

- Document findings, evidence screenshots, and sandbox report in the case notes.
- Update incident status in Defender to “Resolved – Remediated.”

### **Organizational / HR Actions**

#### **User Notification**

- Inform the affected users that an email containing a suspicious URL was automatically removed by Microsoft Defender.
- Advise them **not to click** on links from unknown or unexpected senders.

#### **Awareness Reminder**

- Send a short awareness message to all staff (or targeted group) reminding them:
  - Always check sender address carefully.
  - Hover over links before clicking.
  - Report suspicious emails using the “Report Phishing” button or SOC mailbox.

#### **Password / Account Safety**

- If any user confirms they clicked the link or entered credentials:
  - Instruct them to **change their password immediately**.
  - Verify **MFA is enabled** for the account.
  - Notify IT/SOC to check for unusual login activity.

#### **Training / Follow-up**

- HR or Security Awareness team may include this case as an example in the next phishing-awareness training.
- Emphasize how automated protection (Defender) helped prevent potential credential theft.

#### **Business Impact**

Medium - Suspicious phishing link detected and removed before user interaction. No data loss or business disruption observed.

#### **Quick Hunting Snippets**

search in (EmailEvents)

NetworkMessageId in ("51d71feb-9c9d-4344-aefb-08de0a9b493c")

| take 100

Search keyword: link.mail.beehiiv.com

Filter Delivery action: *Delivered and Removed*

Export list of recipients and message IDs for tracking

## Result / Assessment

### Detection Outcome:

Microsoft Defender automatically detected and removed emails containing a potentially malicious redirect URL after delivery.

### Analysis Result:

The suspicious link (link.mail.beehiiv.com/...) was analyzed in CyberChef and produced unreadable binary data — no clear text or final domain revealed.

Redirect tracing confirmed that the URL uses an encoded token, most likely a **tracking or redirection mechanism**.

No signs of active infection or credential submission detected at this stage.

### Assessment:

**Suspicious / Under Review** — possible *phishing via encoded link* (MITRE: T1566.002).

Classified as **Medium severity** until sandbox or OSINT validation confirms whether the final redirect domain hosts malicious content.

### Current Status:

- ✅ All affected emails removed.
- 🔍 No confirmed user compromise.
- 🕵️ Monitoring and sandbox verification still ongoing.

### Next Action:

Keep incident open pending sandbox report. If the final redirect host is benign (newsletter), mark as *False Positive* and close; if phishing or malware, escalate to Tier-2 for deeper investigation.

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Evidence and Response (7)

Recommended actions (20)

Summary

Alerts

Play attack story

Oct 13, 2025 11:15 PM

In progress

Email messages containing malicious URL removed after delivery

Ahmet Yavas

2 Mailboxes

Incident graph

Layout

Group similar node

ahmet.yavas@csvisor.de

ahmet.yava:

All the A.I. Stories You Miss From This Week

link.mail.beehiv.com

RECOMMENDATIONS

Phishing Incident response playbook

View phishing investigation and response recommended steps for this incident

Open phishing playbook

Incident details

Assigned to: Unassigned

Incident ID: 268

Classification: Not set

Categories: Initial access

First activity: Oct 13, 2025 11:15:10 PM

Last activity: Oct 13, 2025 11:17:10 PM

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Recommended actions (20)

Summary

Incident graph

Layout

Group similar nodes

ahmet.yavas@csvisor.de

ahmet.yava:

All the A.I. Stories You Miss From This Week

link.mail.beehiv.com

Open url page

Original URL

https://link.mail.beehiv.com/ss/c/u001J9xsYZJOg/GGEQWc6GCPUDjicVwQH-Z5qJf-ZxIOx4Q6DR8LPHKHcRVL\_e2EURste4-y7xNBH07upiG2RV55wEISQkr1fK5Rm3NANp18udCnN-U7B8jfeHnrUwX8RQZcovMzA1MLwkwWcmbZ0P\_uocfpm1Tgfu2GbfvIoUmoISN1s\_Y9D6jDbYZvA7m9uQOmnc5KaYq4HAz0L7g4lypG0aZCwGZRLXVVD-uW53MML5kryXrp1HgfZdYLZt15MZG-mpIQ5zdB2XA6BOGuntRsoWh/wGwnG1MrKhxeqIV71FM8x1BWSM4sV05vZQ3fITV6UD3eM\_sl0nU0pWCHym5a7yvb84.7n5goDWWkE5ylexV55b1duJFUSKvwBDxJlB/4kp/1u-VBcxT565PHtIBbNZcjQ/h2/h001.-J4UrMN5loK5m5-Sk6kAOoGwDai1trJJAf-c-726jzk

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ahmet.yavas@csvisor.de

All the A.I. Stories You Miss From This Week

All evidence

Export

7 items

Entity

All the A.I. Stories You Miss From This Week

BodyFingerprintBin1:{"3764384468"} and P2SenderDomain("mail.beehiv.com") and...

BodyFingerprintBin1:{"3764384468"} and SenderIp("159.183.159.49") and ContentT...

https://link.mail.beehiv.com/ss/c/u001J9xsYZJOg/GGEQWc6GCPUDjicVwQH-Z5qJf...

Subject:("All the A.I. Stories You Miss From This Week") and SenderIp("159.183.159.4...

NormalizedUrl("https://link.mail.beehiv.com/ss/c/u001J9xsYZJOg/GGEQWc6GCPUDjicVwQH-Z5qJf...

Subject:("All the A.I. Stories You Miss From This Week") and P2SenderDomain("mail...

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Advanced hunting

Selected workspace: mssentinel

Help resources

New query

Run query

Last 7 days

Save

Share link

Create summary rule

Query

1 search in (EmailEvents)

2 NetworkMessageId in ("51d71feb-9c9d-4344-aefb-88de0a9b493c")

3 take 100

Getting started

Results

Query history

7 items

Search

Customize columns

| Time                                             | Query                                                                                                                                                                                                                                                        | Query time | State     |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| <input type="checkbox"/> Oct 15, 2025 1:56:12 PM | DeviceProcessEvents<br>  where InitiatingProcessFileName == "svchost.exe" or ProcessCommandLine contains "njsrnt"<br>  project Timestamp, DeviceName, ProcessCommandLine, InitiatingProcessFileName, AccountName<br>  order by Timestamp desc                | 0.04s      | Completed |
| <input type="checkbox"/> Oct 15, 2025 1:55:52 PM | DeviceEvents<br>  where ActionType == "AntivirusDetection"<br>  where AdditionalFields contains "njsrnt" or FileName == "svchost.exe"<br>  project Timestamp, DeviceName, FileName, SHA256, InitiatingProcessFileName, ReportId<br>  order by Timestamp desc | 0.04s      | Completed |
| <input type="checkbox"/> Oct 15, 2025 1:55:37 PM | where FileName == "svchost.exe" or SHA256 == "known_njsrnt_hash"<br>  project Timestamp, DeviceName, FolderPath, InitiatingProcessFileName, InitiatingProcess<br>  order by Timestamp desc                                                                   | 0          | Failed    |

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Group similar nodes

Communication

Association

Email messages containing malicious URL re...

All the AI Stories You Miss From This Wee...

Export

Open in advanced hunting

Show empty columns

1 item

Search

00:00.109

Low

Share

| Timestamp                                         | AlertId                                                        | Title                      | Category      | Severity      | ServiceSource              |
|---------------------------------------------------|----------------------------------------------------------------|----------------------------|---------------|---------------|----------------------------|
| <input type="checkbox"/> Oct 13, 2025 11:17:10 PM | fa254b2029-0c41-...                                            | Email messages containi... | InitialAccess | Informational | Microsoft Defender for ... |
| Timestamp                                         | Oct 13, 2025 11:17:10 PM                                       |                            |               |               |                            |
| AlertId                                           | fa254b2029-0c41-07f2-4400-08de0a9cdea9                         |                            |               |               |                            |
| Title                                             | Email messages containing malicious URL removed after delivery |                            |               |               |                            |
| Category                                          | InitialAccess                                                  |                            |               |               |                            |
| Severity                                          | Informational                                                  |                            |               |               |                            |
| ServiceSource                                     | Microsoft Defender for Office 365                              |                            |               |               |                            |
| DetectionSource                                   | Microsoft Defender for Office 365                              |                            |               |               |                            |

link.mail.beehiv.com

At least 2 detected files communicating with this domain

Reanalyze

Similar

More

0 / 95

Community Score

link.mail.beehiv.com

beeihiv.com

Email, Marketing/Merchandising (alphaMountain.ai)

trackers

general business

top-100K

Registrar

Cloudflare, Inc.

Creation Date

5 years ago

Last Analysis Date

1 hour ago

DETECTION

DETAILS

RELATIONS

COMMUNITY 14

Contained in Graphs (1)

sandaruwijayasena

link.mail.beehiv.com

2024-01-26 09:59:54

Voting details (12)

ryanbeehiv 3 months ago +1

Deone01 7 months ago +1

colbybh 10 months ago +1

TKadmin 1 year ago -1

Meeena 1 year ago -1

124394 1 year ago +1





TECHNIQUES

- Spearphishing Link** ▲
- Spearphishing via Service
- Spearphishing Voice
- Replication Through Removable Media
- Supply Chain Compromise ▾
- Trusted Relationship ▾
- Valid Accounts ▾
- Wi-Fi Networks
- Execution ▾
- Persistence ▾
- Privilege Escalation ▾
- Defense Evasion ▾
- Credential Access ▾

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# Phishing: Spearphishing Link

Other sub-techniques of Phishing (4) ▾

Adversaries may send spearphishing emails with a malicious link in an attempt to gain access to victim systems. Spearphishing with a link is a specific variant of spearphishing. It is different from other forms of spearphishing in that it employs the use of links to download malware contained in email, instead of attaching malicious files to the email itself, to avoid defenses that may inspect email attachments. Spearphishing may also involve social engineering techniques, such as posing as a trusted source.

All forms of spearphishing are electronically delivered social engineering targeted at a specific individual, company, or industry. In this case, the malicious emails contain links. Generally, the links will be accompanied by social engineering text and require the user to actively click or copy and paste a URL into a browser, leveraging *User Execution*. The visited website may compromise the web browser using an exploit, or the user will be prompted to download applications, documents, zip files, or even executables depending on the pretext for the email in the first place.

Adversaries may also include links that are intended to interact directly with an email reader,

ID: T1566.002

Sub-technique of: T1566

① **Tactic:** [Initial Access](#)

① **Platforms:** Identity Provider, Linux, Office Suite, SaaS, Windows, macOS

**Contributors:** Jeff Sakowicz, Microsoft Identity Developer Platform Services (IDPM Services); Kobi Haimovich, CardinalOps; Mark Wee; Menachem Goldstein; Philip Winther; Saisha Agrawal, Microsoft Threat Intelligent Center (MSTIC); Shailesh Tiwary (Indian Army)

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